ElectViz Election Data Visualization on for Media

Transforming raw election data into actionable insights through interactive Power BI visualisation

Project Vision & Objectives

Party Performance Analysis

Compare political party performance across different states and election years to identify patterns and shifts in voter preferences

Voter Turnout Patterns

Study participation trends and engagement levels across constituencies to understand democratic involvement

Election Trend Identification

Uncover key trends and emerging political influences that shape electoral outcomes and predict future patterns

Dataset Structure & Components

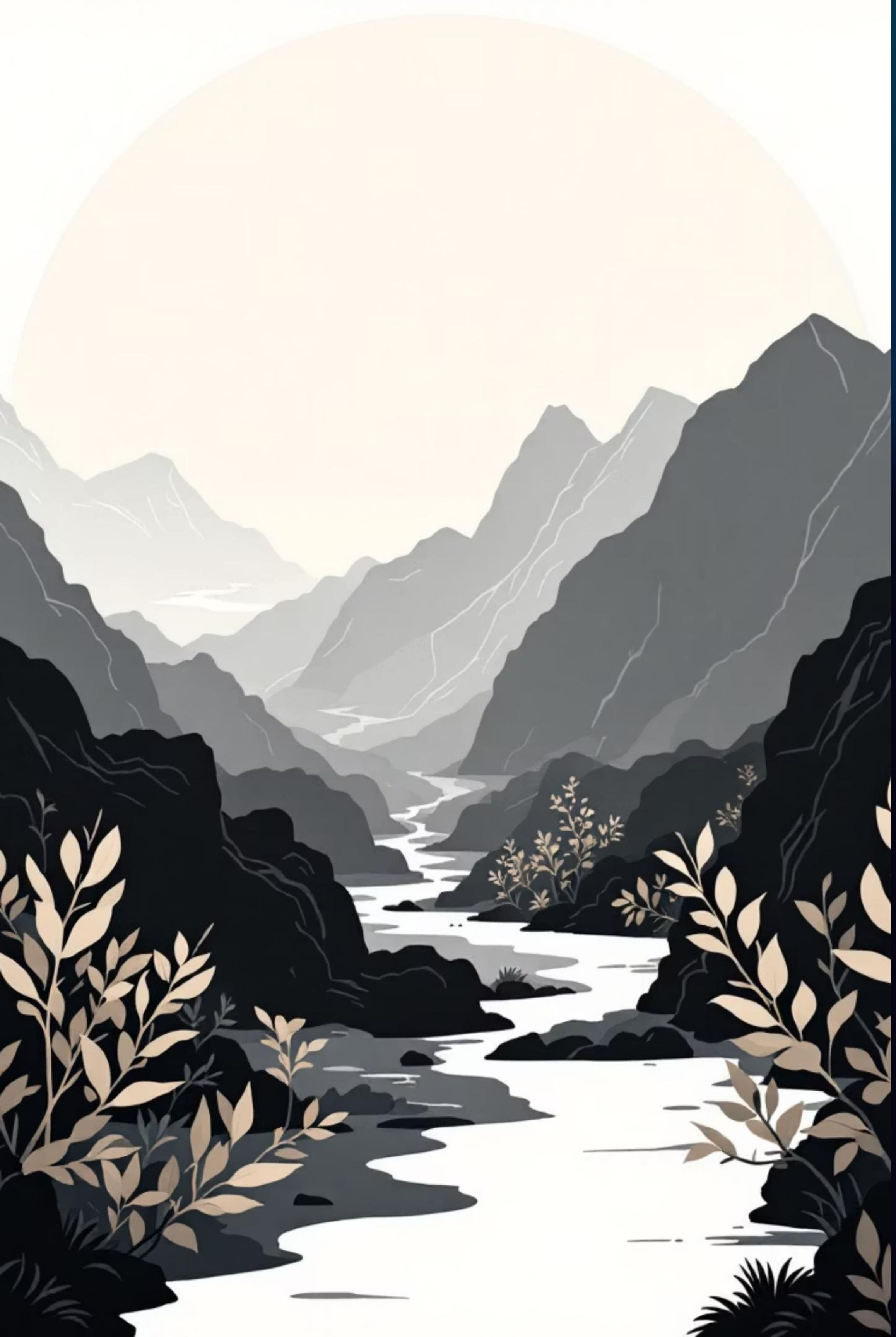
Core Data Elements

- **State:** Geographic electoral regions
- **Constituency:** Individual voting areas
- **Candidate:** Contesting individuals
- **Party:** Political affiliations
- **Votes:** Total votes received
- **Year:** Election cycle timestamps

Data Quality Measures

Comprehensive data preparation ensures accuracy and reliability for meaningful analysis.

- Missing record identification and removal
- Duplicate entry detection and elimination
- Data type validation and standardisation
- Consistency checks across all fields



Data Transformation Process

Data Import

Raw election data loaded into Power Query for processing

Cleaning & Validation

Removing duplicates, handling missing values, standardising formats

Calculated Metrics

DAX measures created for vote share, turnout percentage, seats won

Model Design

Star schema implementation with fact and dimension tables

Star Schema Data Model

Model Architecture

A well-structured **Star Schema** ensures optimal query performance and accurate analysis across multiple dimensions.

Unique keys establish relationships between tables, enabling seamless data integration and complex calculations.

Fact Table

Election Results: Contains transactional data including votes, candidates, and outcomes

Dimension Tables

- Party dimension – Political affiliations
- Candidate dimension – Contestant details
- State dimension – Geographic attributes
- Year dimension – Temporal analysis

Interactive Dashboard Features

Dynamic Slicers

Year, State, and Party filters enable users to drill down into specific election segments and customise their analysis view

Key Metric Cards

Prominent display of Total Votes, Winning Party, and Voter Turnout statistics provides immediate insights at a glance

Visual Analytics

Bar charts, pie charts, trend lines, and geographical maps present election data through multiple analytical lenses

Visual Elements Portfolio

Bar Charts

Compare party performance and constituency results side by side

Pie Charts

Visualise vote share distribution and party proportions

Trend Lines

Track voter turnout and party performance over multiple election cycles

Map Visuals

Display geographical patterns and regional political dominance

Each visual element is strategically chosen to communicate specific insights, ensuring the dashboard tells a complete electoral story.

Key Insights Discovered

Regional Political Dominance

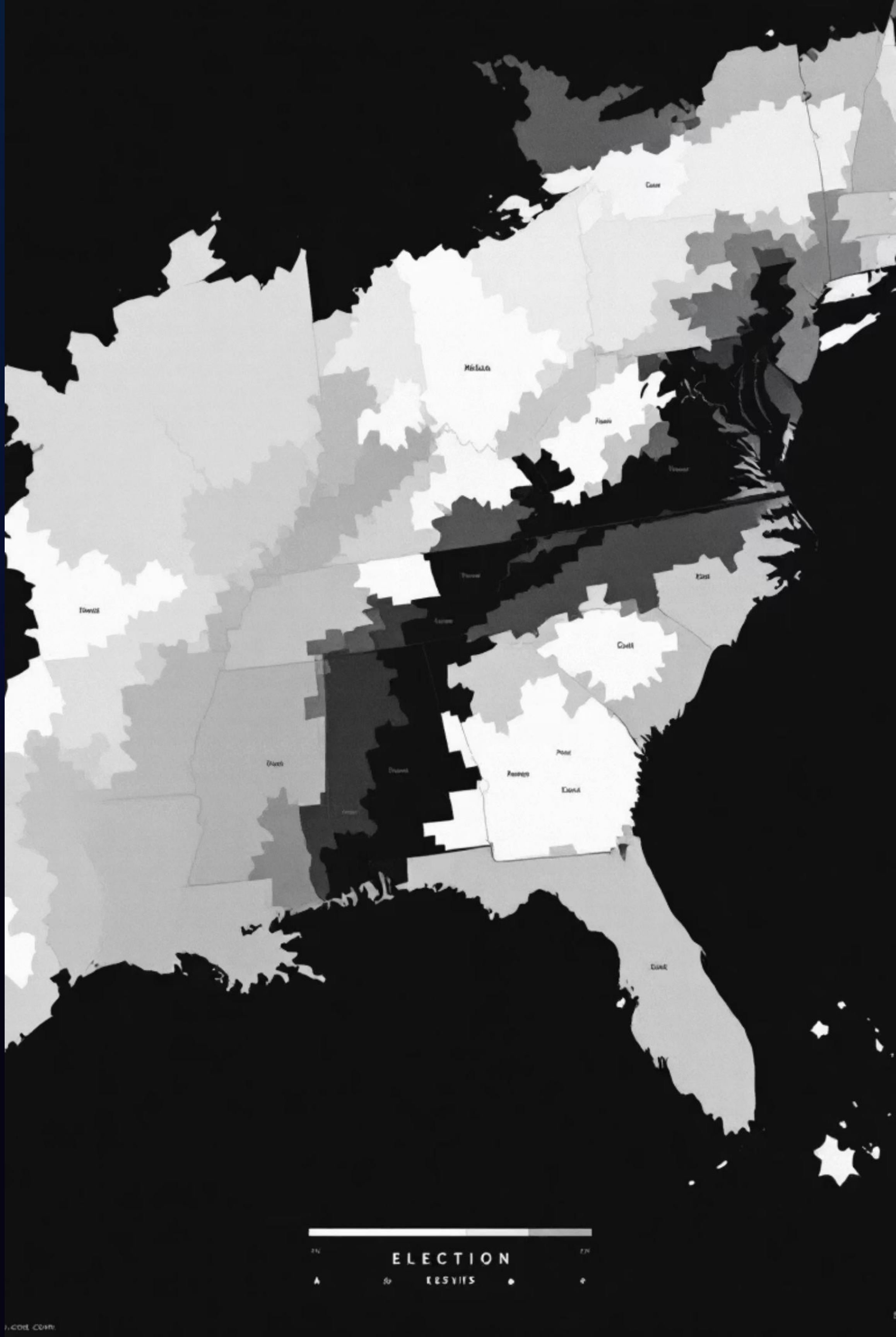
Major parties demonstrate clear geographic strongholds, with consistent performance patterns across specific states revealing deep-rooted voter loyalty and regional political identities.

Voter Turnout Variations

Significant fluctuations in participation rates across states and years indicate varying levels of political engagement, influenced by local issues and national campaign intensity.

Emerging Political Forces

New party influences are gaining traction in select constituencies, suggesting shifting voter preferences and the potential for future electoral realignment in these areas.



The Power of Data-Driven Democracy

Transforming Electoral Analysis

This project demonstrates how **Power BI** transforms complex election datasets into clear, actionable intelligence.

By combining robust data modelling, DAX calculations, and interactive visualisations, raw electoral information becomes a powerful tool for:

- Strategic political planning
- Voter behaviour understanding
- Policy impact assessment
- Democratic process transparency

Project Credentials

100%

Power BI Native

Built entirely within Microsoft Power BI ecosystem

4

Dimension Tables

Comprehensive star schema data model

6+

Visual Types

Diverse chart and map visualisations

Developed by

Charan Teja

Thank you !